Exercise 3.1: Investigating Cellular Network Parameters Using Service Mode

Objective
The objective of this experiment is to investigate and analyze key cellular network parameters using the Field Test Mode of an Apple iPhone. The study aims to observe signal strength, signal quality, and cell-related parameters and understand how these affect network performance and reliability.

Device and Tools Used
Device: Apple iPhone
Operating System: iOS
Network Operator: DNA (Finland)
Service Mode: iOS Field Test Mode (*3001#12345#*)

Observed Parameters
RSRP (Signal Strength), RSRQ (Signal Quality), SINR (Interference), PCI (Cell ID), Network Technology (LTE).

Measurement Summary
Two LTE measurements were recorded. RSRP values (-85 dBm and -84 dBm) indicate good signal strength, while SINR values (-20 dB) indicate high interference. A change in PCI suggests cell reselection.

Analysis
Despite good signal strength, poor SINR shows that interference and network congestion significantly affect performance. This highlights that signal quality parameters are often more critical than signal strength alone.

Conclusion
The experiment demonstrates that strong signal strength does not guarantee high-quality network performance. Both signal strength and signal quality parameters must be considered when evaluating cellular networks.

Figure 1: iPhone Field Test Mode – Measurement 1

10.05

◀ Phone

Dashboard

Carrier:	DNA
Network Capabilities:	SA+NSA
TAC:	37922
PLMN:	244 12
Phone Number:	+358417165753

LTE

PCI	452
RSRP	-85 dBm
RSRQ	-10 dB
SINR0	-20.0 dB
SINR1	-20.0 dB

Dashboard
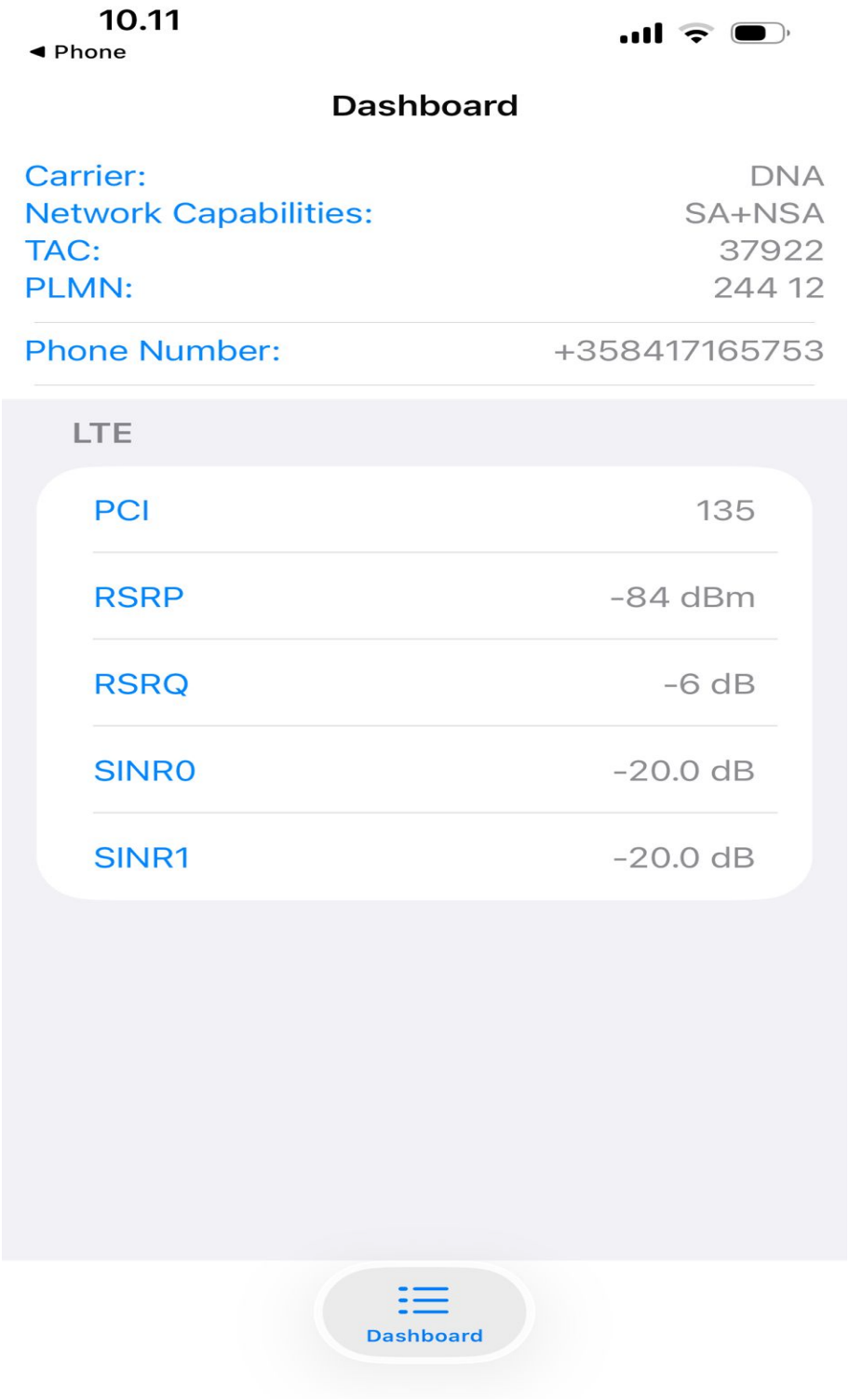
Figure 2: iPhone Field Test Mode – Measurement 2